BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Introduction to Data Science

Course Overview & Introduction

- *The slides presented here are obtained from the authors of the books and from various other contributors. I hereby acknowledge all the contributors for their material and inputs.*
- *We have added and modified slides to suit the requirements of the course.*

Course Objectives

CO1 Gain basic understanding of the role of Data Science in various scenarios in the real-world of business, industry and government.

CO2 Understand various roles and stages in a Data Science Project and ethical issues to be considered.

CO3 Explore the processes, tools and technologies for collection and analysis of structured and unstructured data.

CO4 Appreciate the importance of techniques like data visualization, storytelling with data for the effective presentations of the outcomes with the stakeholders.

CO5 Understand techniques of preparing real-world data for data analytics.

CO6 Implement data analytic techniques for discovering interesting patterns from data.

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Text Books

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflic; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers

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Modular Structure

No	Title of the Module
M1	Fundamentals of Data Science
M2	Data Models and Pipelines
M3	Data wrangling
M4	Feature Engineering
M5	Classification and Prediction
M6	Association Analysis
M7	Clustering
M8	Anomaly Detection
M9	Storytelling with Data
M10	Ethics for Data Science

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Evaluation Scheme

No	Name	Type	Weight
1.	Quiz-I	Online	5%
	Quiz-II	Online	5%
	Assignment-1	Group	10%
	Assignment-2	Group	10%
2.	Mid-Semester Test		25%
3.	Comprehensive Exam		45%

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Data Science Defined

What Is Data Science?

- Data science uncovers actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning.
- Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI), and machine learning with specific subject matter expertise

Why data science?

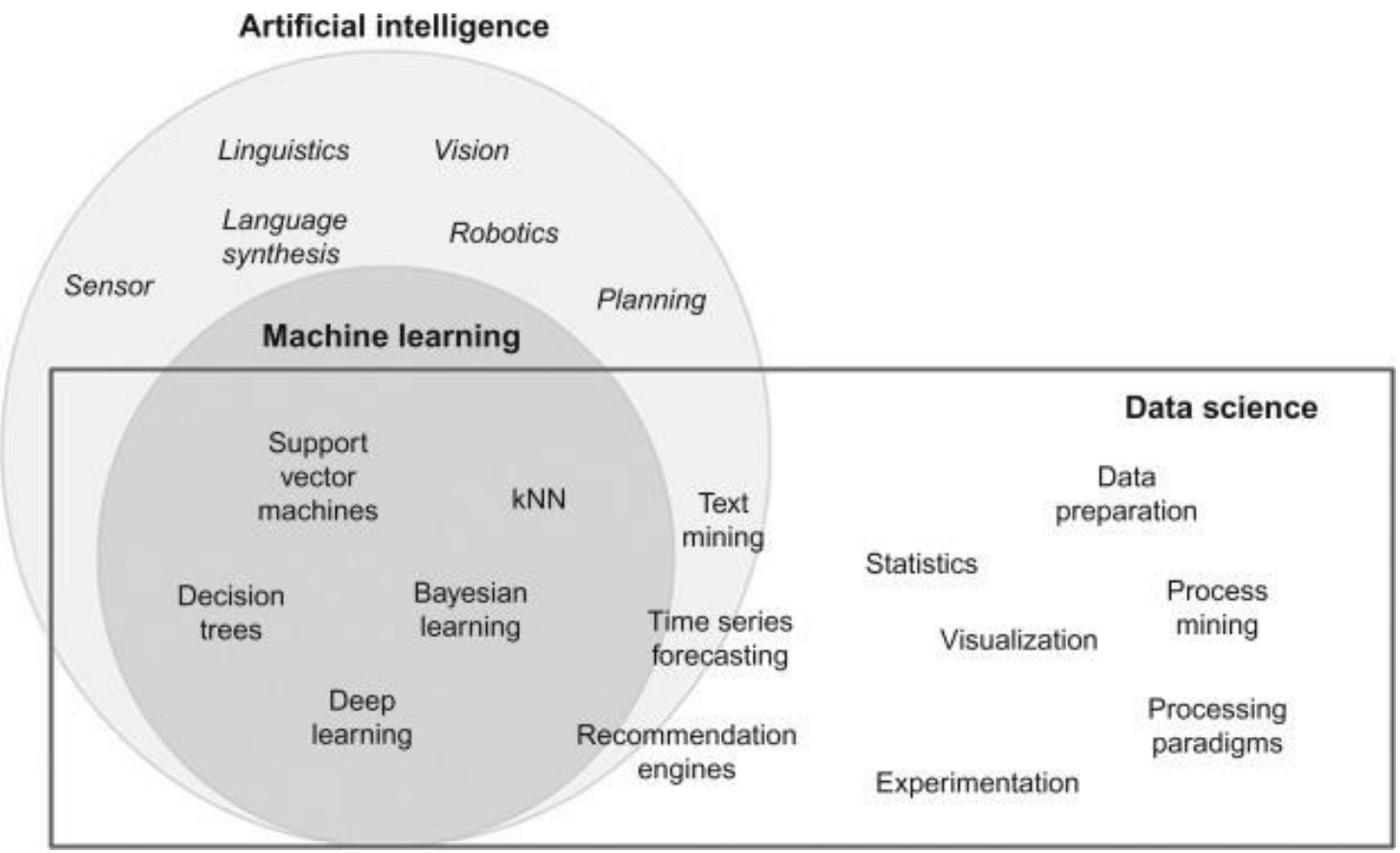
- The accelerating volume of data sources, and subsequently data, has made data science as one of the fastest growing field across every industry.

The term "data scientist" was coined as recently as 2008 when companies realized the need for data professionals who are skilled in organizing and analyzing massive amounts of data.

- *https://ischoolonline.berkeley.edu/data-science/what-is-data-science/*

https://www.ibm.com/topics/data-science

Data Science, AI, and ML



https://www.sciencedirect.com/topics/physics-and-astronomy/artificial-intelligence

Data Science & Machine Learning

According to Tom M. Mitchell, Chair of Machine Learning at Carnegie Mellon University and author of the book *Machine Learning* (McGraw-Hill),

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with the experience E.

We now have a set of objects to define machine learning:

Task (T), Experience (E), and Performance (P)

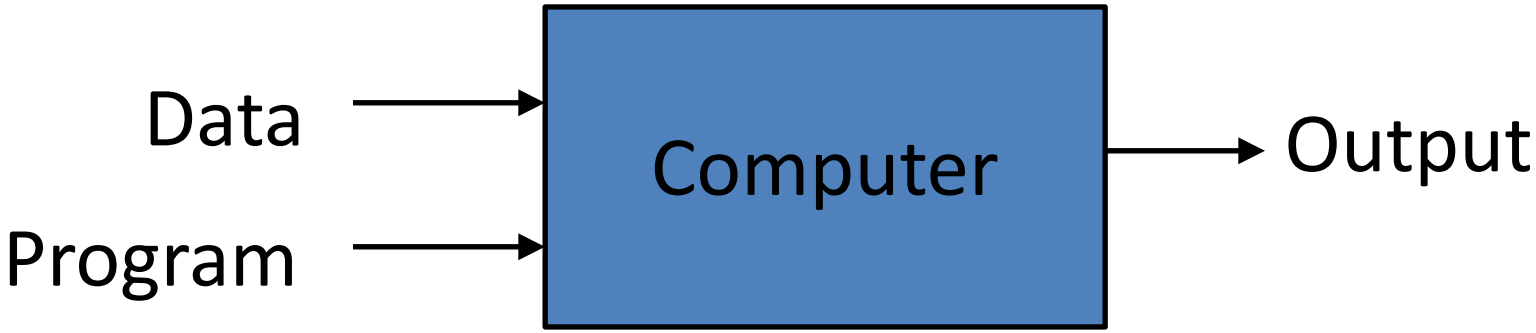
With a computer running a set of tasks, the experience should be leading to performance increases (to satisfy the definition)

Data Science tasks are executed successfully with help of machine learning

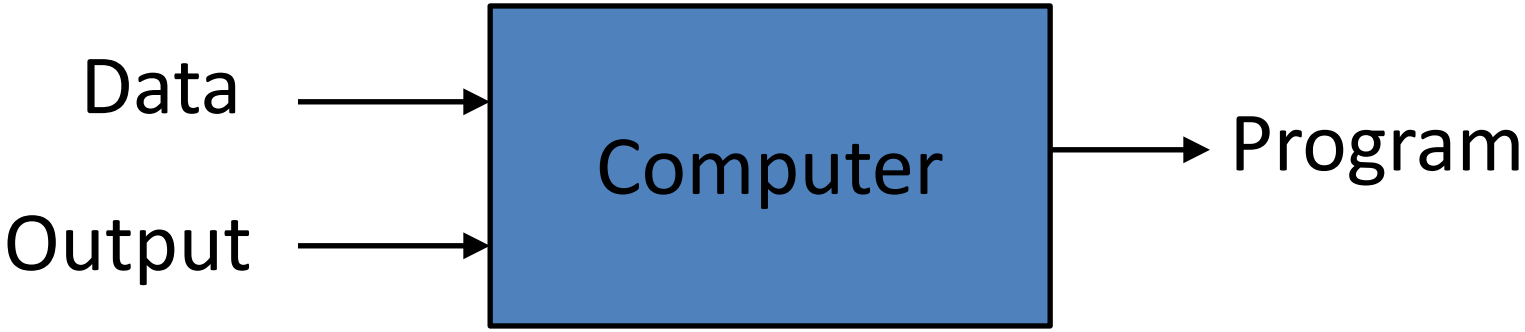
Machine Learning: Hands-on for Developers and Technical Professionals by Jason Bell John Wiley & Sons

Traditional Programming vs ML

Traditional Programming



Machine Learning



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Types of Learning

- **Supervised (inductive) learning**
  - Given: training data, desired outputs (labels)
- **Unsupervised learning**
  - Given: training data (without desired outputs)
- **Semi-supervised learning**
  - Given: training data + a few desired outputs
- **Reinforcement learning**
  - Given: rewards from sequence of actions

Slide Credit: Eric Eaton

Supervised, Unsupervised and Reinforcement Learning

Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method

https://intellipaat.com/blog/supervised-learning-vs-unsupervised-learning-vs-reinforcement-learning/

Query Response vs Pattern Discovery

Query Responses	Pattern Discovery
Applies to Databases	Applies to diverse Repositories
Mostly SQL or other query languages	Usually Machine Learning
Definitive	Approximate/Probabilistic

Data in Enterprise

Data in Enterprise

Can be
- Structured
- Semi-structured
- Unstructured

Can have varying ROB (return on byte)

May require
- ACID property
- BASE may be fine

Can have varying velocity
- Stream
- Batch

Databases

Can be
- SQL based RDBMS (mot business transactions)
  - Support for ACID
- NoSQL databases
  - Mostly BASE

SQL based RDBMS	NoSQL based DBMS

• Transactional/Operational DBMS
  • Designed by ER modeling
  • Serves OLTP
• Data Warehouse
  • Designed by Star/Snowflake modeling
  • Serves OLAP
	• Key-Value Stores
	• Document Databases
	• Column Family Stores
	• Graph Databases

Why go beyond RDBMS?

Its strengths are also its weakness

	Strength	Weakness
Schema	Clearly defines data and relationship; ensures data quality and integrity.	Not suitable for semi-structured data.
ACID	Guarantees the correctness of the operations and the durability of the data.	Makes it very difficult to scale
SQL	One language for all data and all RDBMS.	Impedes rapid application development (RAD) due to the mismatch between SQL and the application languages.

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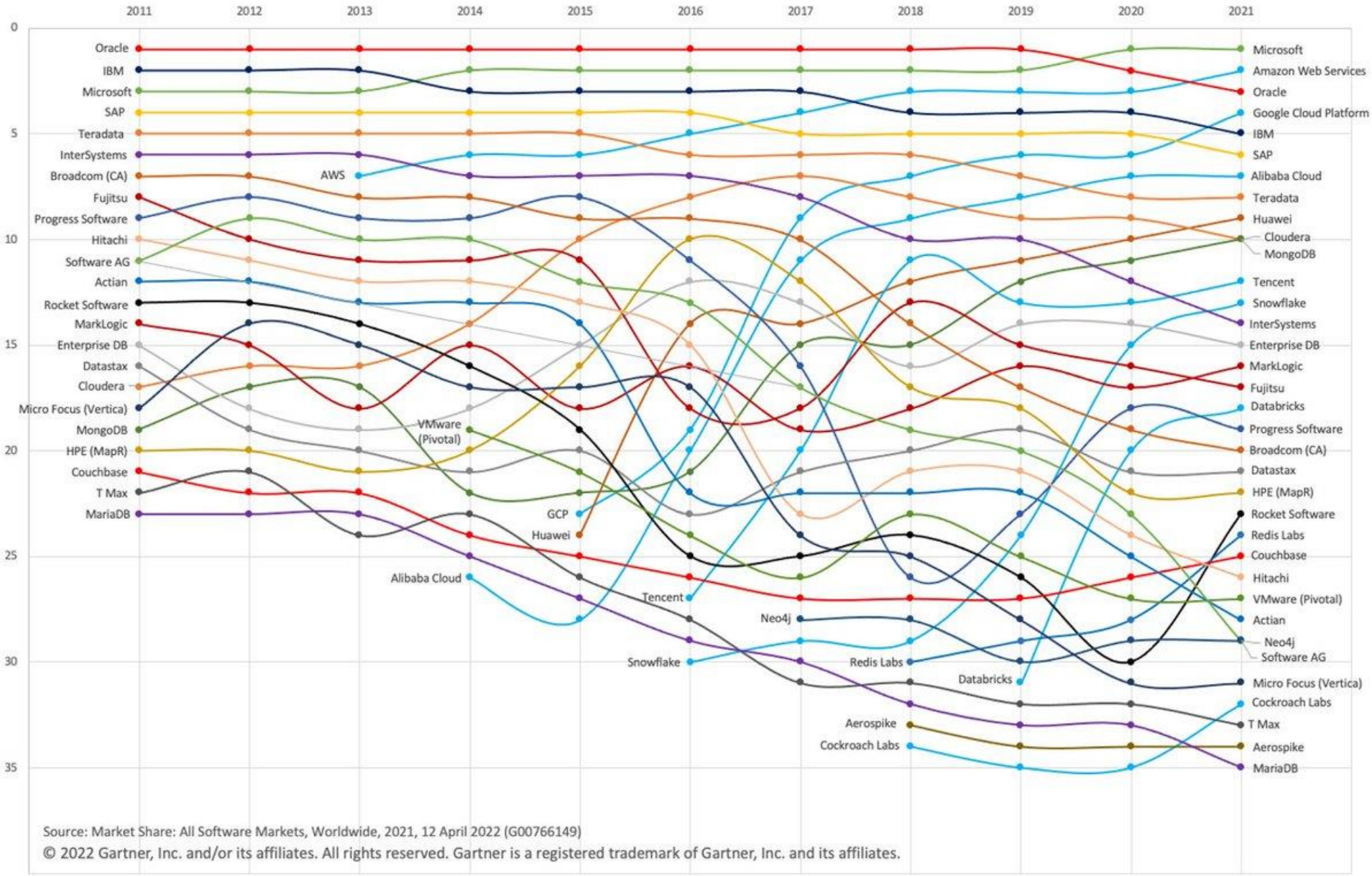
Gartner DBMS Market Share Ranks: 2011-2021



Source: Market Share: All Software Markets, Worldwide, 2021, 12 April 2022 (G00766149)

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https://twitter.com/ARonthal/status/1514595630072619014 https://www.techrepublic.com/article/new-gartner-report-shows-massive-growth-database-market-fueled-cloud/

Diverse kinds of Data

Besides relational database data (from operational or analytical systems), there are many other kinds of data that have diverse forms and structures and different semantic meanings.

Examples of data can be :

- time-related or sequence data (e.g., historical records, stock exchange data, and time-series and biological sequence data),
- data streams (e.g., video surveillance and sensor data, which are continuously transmitted),
- spatial data (e.g., maps),
- engineering design data (e.g., the design of buildings, system components, or integrated circuits),
- hypertext and multimedia data (including text, image, video, and audio data),
- graph and networked data (e.g., social and information networks), and
- the Web (a widely distributed information repository).

Diversity of data brings in new challenges such as handling special structures (e.g., sequences, trees, graphs, and networks) and specific semantics (such as ordering, image, audio and video contents, and connectivity)

Amazon perspective:

Data Architecture for Enterprise

https://aws.amazon.com/big-data/datalakes-and-analytics/modern-data-architecture/?nc=sn&loc=2

Data Science Roles

https://www.altexsoft.com/blog/datascience/how-to-structure-data-science-team-key-models-and-roles/

Data Science Activity Examples

Classification: Definition

Given a collection of records (*training set*)

- Each record contains a set of *attributes*, one of the attributes is the *class*.

Find a *model* for class attribute as a function of the values of other attributes.

Goal: previously unseen records should be assigned a class as accurately as possible.

- A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

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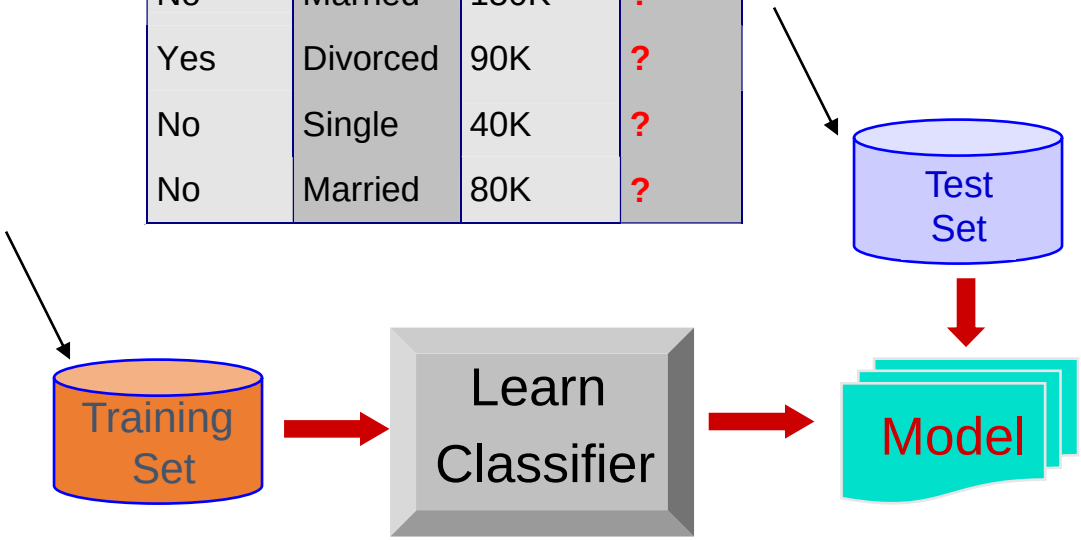
Classification Example

Tid	Refund	Marital Status	Taxable Income	Cheat
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?



Classification: Application 1

Direct Marketing

- Goal: Reduce cost of mailing by *targeting* a set of consumers likely to buy a new cell-phone product.
- Approach:
  - Use the data for a similar product introduced before.
  - We know which customers decided to buy and which decided otherwise. This *{buy, don't buy}* decision forms the *class attribute*.
  - Collect various demographic, lifestyle, and company-interaction related information about all such customers.
    - Type of business, where they stay, how much they earn, etc.
  - Use this information as input attributes to learn a classifier model.

From [Berry & Linoff] Data Mining Techniques, 1997

Classification: Application 2

Fraud Detection

- Goal: Predict fraudulent cases in credit card transactions.
- Approach:
  - Use credit card transactions and the information on its account-holder as attributes.
    - When does a customer buy, what does he buy, how often he pays on time, etc
  - Label past transactions as fraud or fair transactions. This forms the class attribute.
  - Learn a model for the class of the transactions.
  - Use this model to detect fraud by observing credit card transactions on an account.

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Classification: Application 3

Customer Attrition/Churn:

- Goal: To predict whether a customer is likely to be lost to a competitor.
- Approach:
  - Use detailed record of transactions with each of the past and present customers, to find attributes.
    - How often the customer calls, where he calls, what time-of-the day he calls most, his financial status, marital status, etc.
  - Label the customers as loyal or disloyal.
  - Find a model for loyalty.

From [Berry & Linoff] Data Mining Techniques, 1997

Clustering Definition

Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that

- Data points in one cluster are more similar to one another.
- Data points in separate clusters are less similar to one another.

Similarity Measures:

- Euclidean Distance if attributes are continuous.
- Other Problem-specific Measures.

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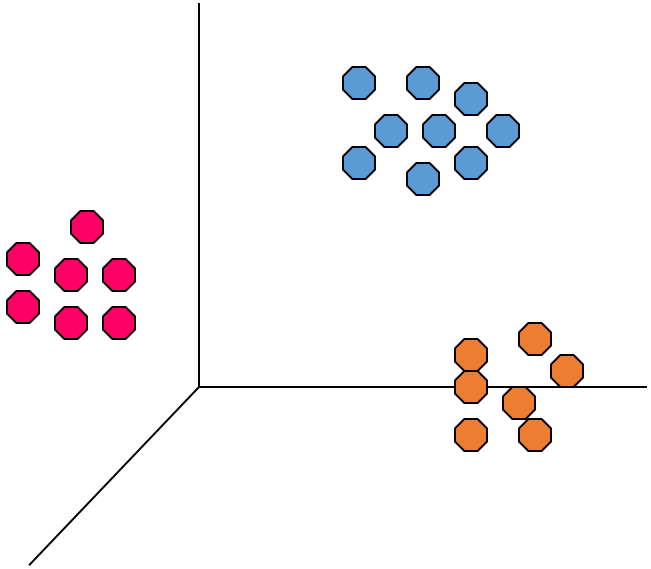
Illustrating Clustering

Intracluster distances are minimized

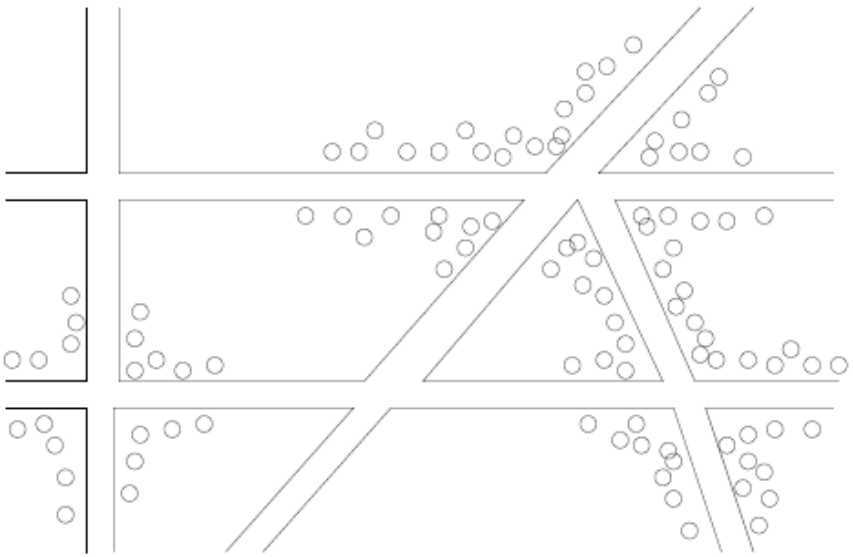
Intercluster distances are maximized

Euclidean Distance Based Clustering in 3-D space



Plotting cholera cases on a map of London

A famous instance of clustering to solve a problem took place long ago in London, and it was done entirely without computers. The physician John Snow, dealing with a Cholera outbreak plotted the cases on a map of the city.



Mining of Massive Datasets, Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman

Clustering: Application 1

Market Segmentation:

- Goal: subdivide a market into distinct subsets of customers where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.
- Approach:
  - Collect different attributes of customers based on their geographical and lifestyle related information.
  - Find clusters of similar customers.
  - Measure the clustering quality by observing buying patterns of customers in same cluster vs. those from different clusters.

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Clustering: Application 2

Document Clustering:

- Goal: To find groups of documents that are similar to each other based on the important terms appearing in them.
- Approach: To identify frequently occurring terms in each document. Form a similarity measure based on the frequencies of different terms. Use it to cluster.
- Gain: Information Retrieval can utilize the clusters to relate a new document or search term to clustered documents.

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Association Rule Discovery: Definition

Given a set of records each of which contain some number of items from a given collection;

- Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

TID	Items
1	Bread, Milk
2	Bread, Diaper, Butter, Beans
3	Milk, Diaper, Butter, Coke
4	Bread, Milk, Diaper, Butter
5	Bread, Milk, Diaper, Coke

Example of Association Rules

{Diaper} → {Butter},
{Milk, Bread} → {Beans, Coke},
{Butter, Bread} → {Milk},

Association Rule Discovery: Application 1

Marketing and Sales Promotion:

- Let the rule discovered be

  {Bagels, … } --> {Potato Chips}

- Potato Chips as consequent => Can be used to determine what should be done to boost its sales.
- Bagels in the antecedent => Can be used to see which products would be affected if the store discontinues selling bagels.
- Bagels in antecedent *and* Potato chips in consequent => Can be used to see what products should be sold with Bagels to promote sale of Potato chips!

Association Rule Discovery: Application

Inventory Management:

- Goal: A consumer appliance repair company wants to anticipate the nature of repairs on its consumer products and keep the service vehicles equipped with right parts to reduce on number of visits to consumer households.
- Approach: Process the data on tools and parts required in previous repairs at different consumer locations and discover the co-occurrence patterns.

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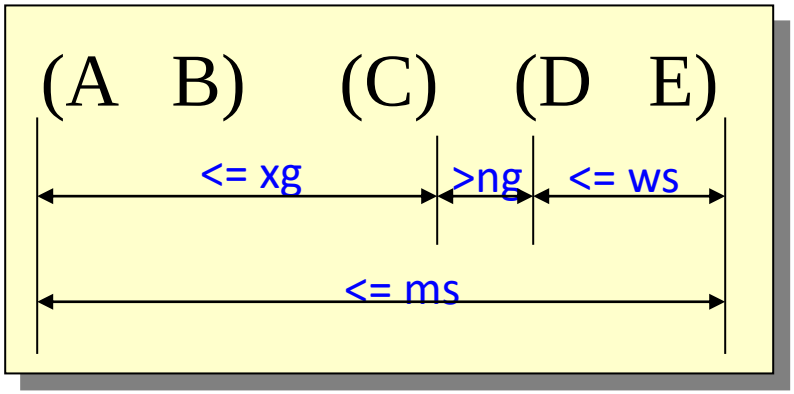
Sequential Pattern Discovery: Definition

Given is a set of *objects*, with each object associated with its own *timeline of events*, find rules that predict strong sequential dependencies among different events.

$$(A \quad B) \quad (C) \longrightarrow (D \quad E)$$

Rules are formed by first discovering patterns. Event occurrences in the patterns are governed by timing constraints.



Timing constraints include maxgap (xg), mingap (ng), windowsize (ws), maxspan (ms)

Sequential Pattern Discovery: Examples

In telecommunications alarm logs,

- (Inverter_Problem Excessive_Line_Current)
  (Rectifier_Alarm) --> (Fire_Alarm)

In point-of-sale transaction sequences,

- Computer Bookstore:
  (Intro_To_Visual_C) (C++_Primer) -->
  (Perl_for_dummies,Tcl_Tk)
- Athletic Apparel Store:
  (Shoes) (Racket, Racketball) --> (Sports_Jacket)

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Prediction/Regression

Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.

Greatly studied in statistics, neural network fields.

Examples:

- Predicting sales amounts of new product based on advertising expenditure.
- Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
- Time series prediction of stock market indices.

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Deviation/Anomaly Detection

Detect significant deviations from normal behavior

Applications:

- Credit Card Fraud Detection
- Network Intrusion Detection

DS Process & Challenges

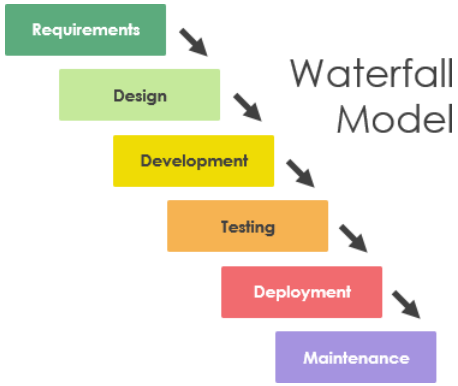
Software Engineering

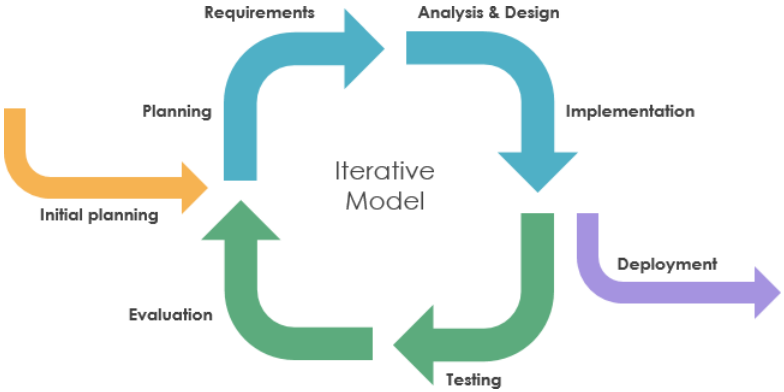
In general,

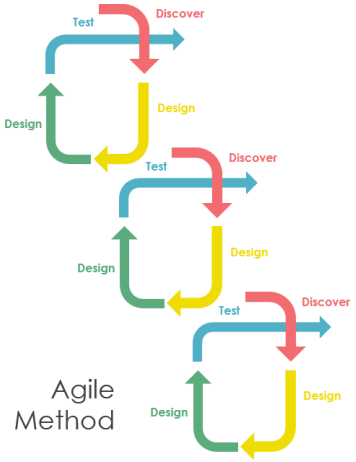
Software engineering is an engineering discipline that is concerned with all aspects of software production.

Software includes computer programs, all associated documentation, and configuration data that are needed for software to work correctly.

Waterfall model, Iterative models, Agile models







DS Process

The standard process involves

1. understanding the problem,
2. preparing the data (samples),
3. developing the model,
4. applying the model on a data set to see how the model may work in real world, and
5. production deployment.

A popular data mining process frameworks is CRISP-DM (Cross Industry Standard Process for Data Mining). This framework was developed by a consortium of companies involved in data mining

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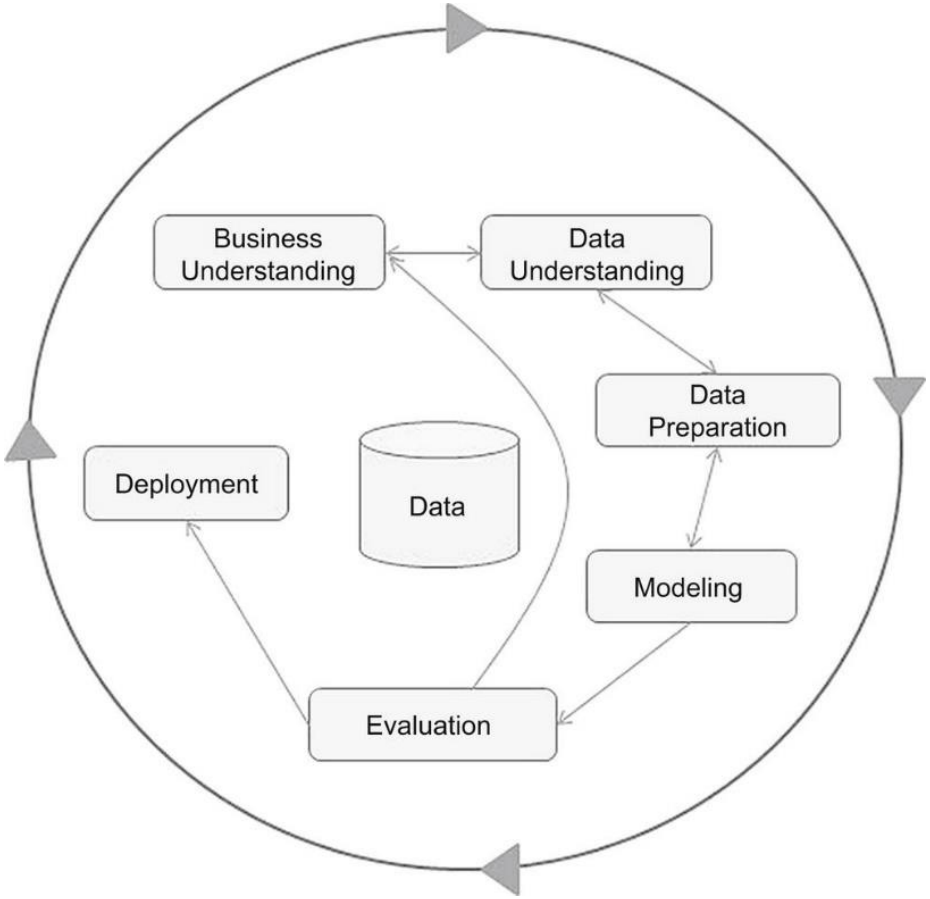
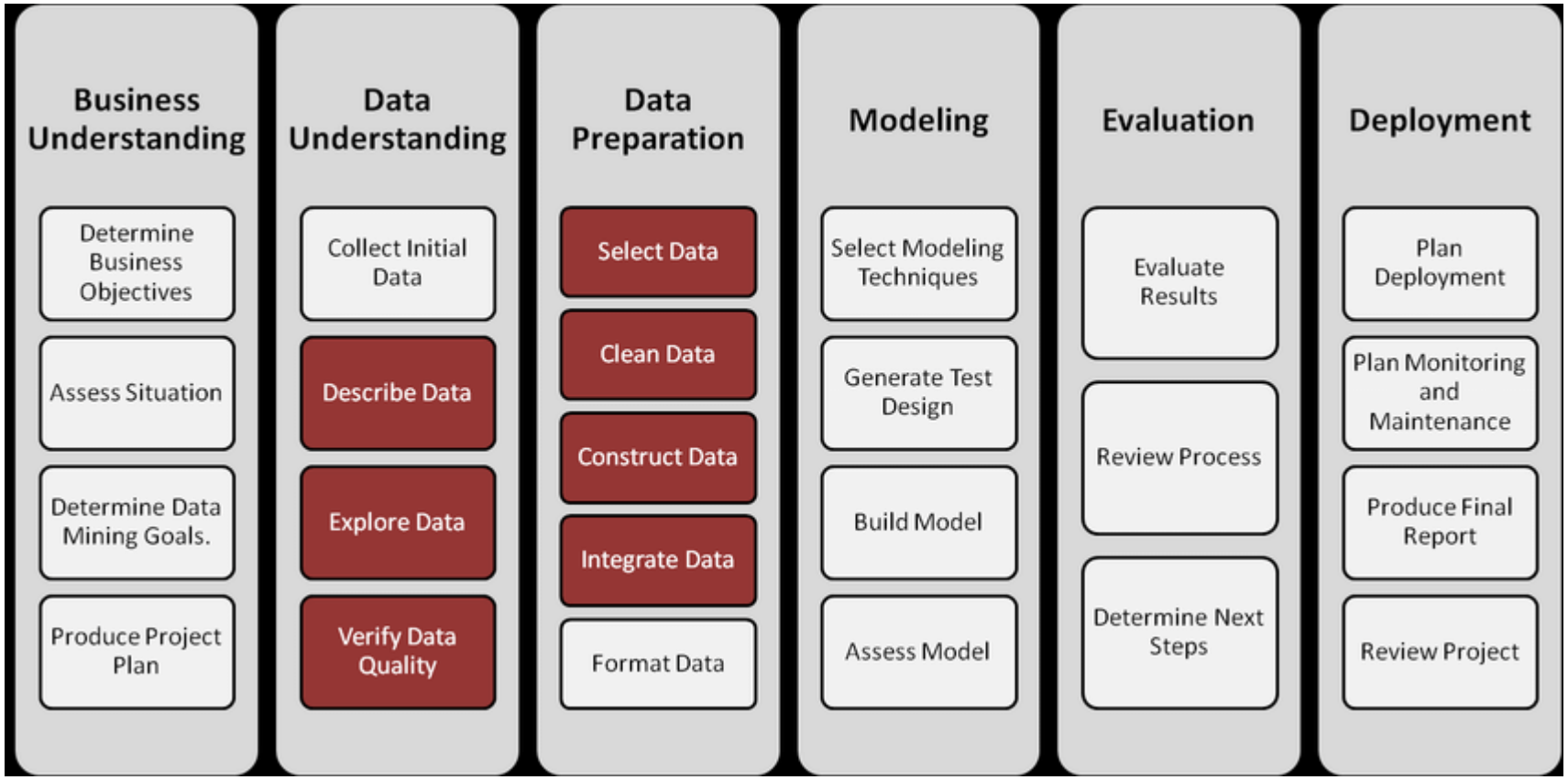
CRISP data mining framework



CRISP is the most popular methodology for analytics, data mining, and data science projects, with 49% share as per 2020 DSPA (Data Science Process Alliance) Poll.

CRISP-DM was conceived in 1996. In 1997 it got underway as a European Union project, led by SPSS, Teradata, Daimler AG, NCR Corporation and OHRA.

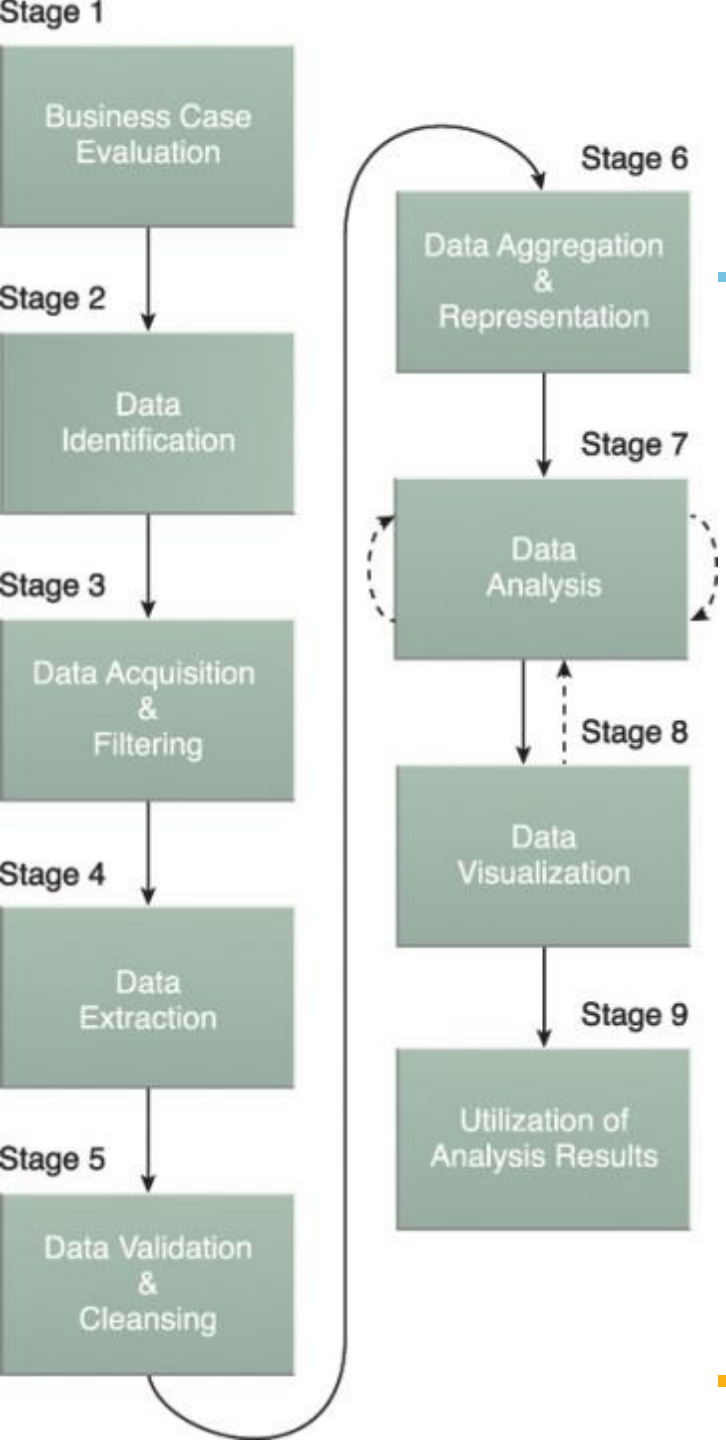
CRISP Phases & Tasks



More on CRISP-DM

https://www.datascience-pm.com/wp-content/uploads/2021/08/CRISP-DM-for-Data-Science.pdf

Big Data Analytics Lifecycle



Business Case Evaluation
requires that a business case be created, assessed and approved prior to proceeding with the actual hands-on analysis tasks.

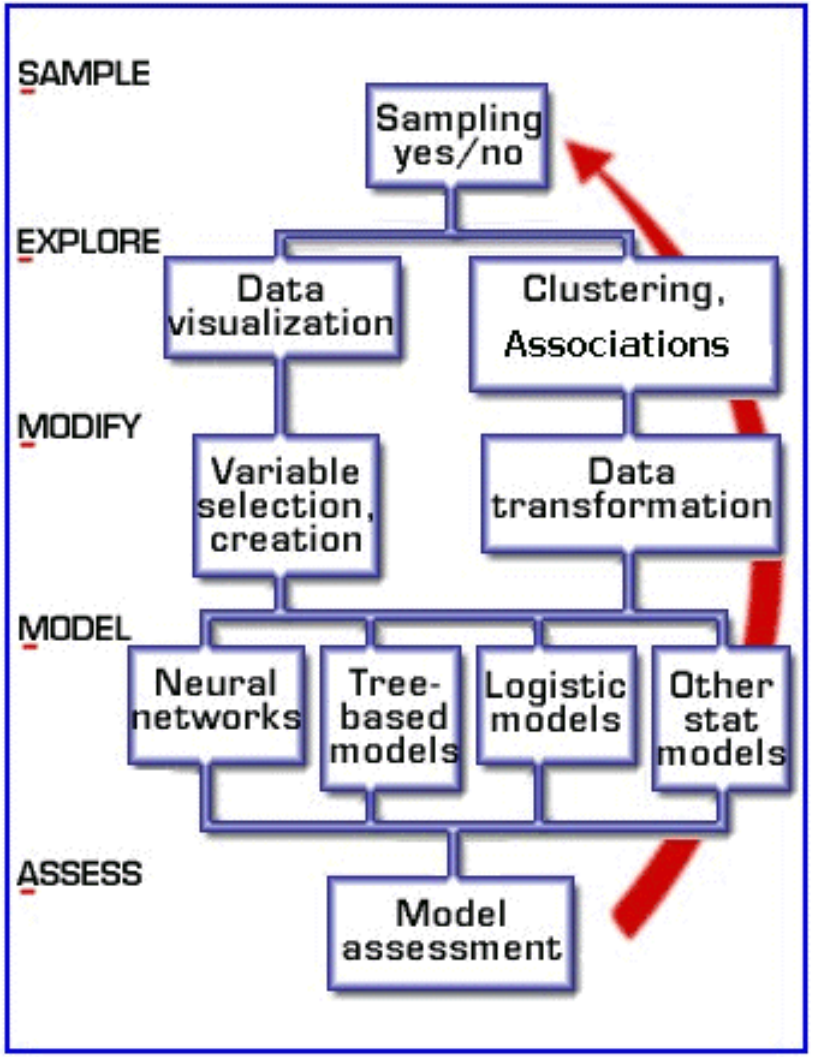
Data Acquisition and Filtering
the data is gathered from all of the identified data sources & then subjected to automated filtering for the removal of corrupt data or irrelevant data.

Data Validation and Cleansing
data input into Big Data analyses can be unstructured without any indication of validity.

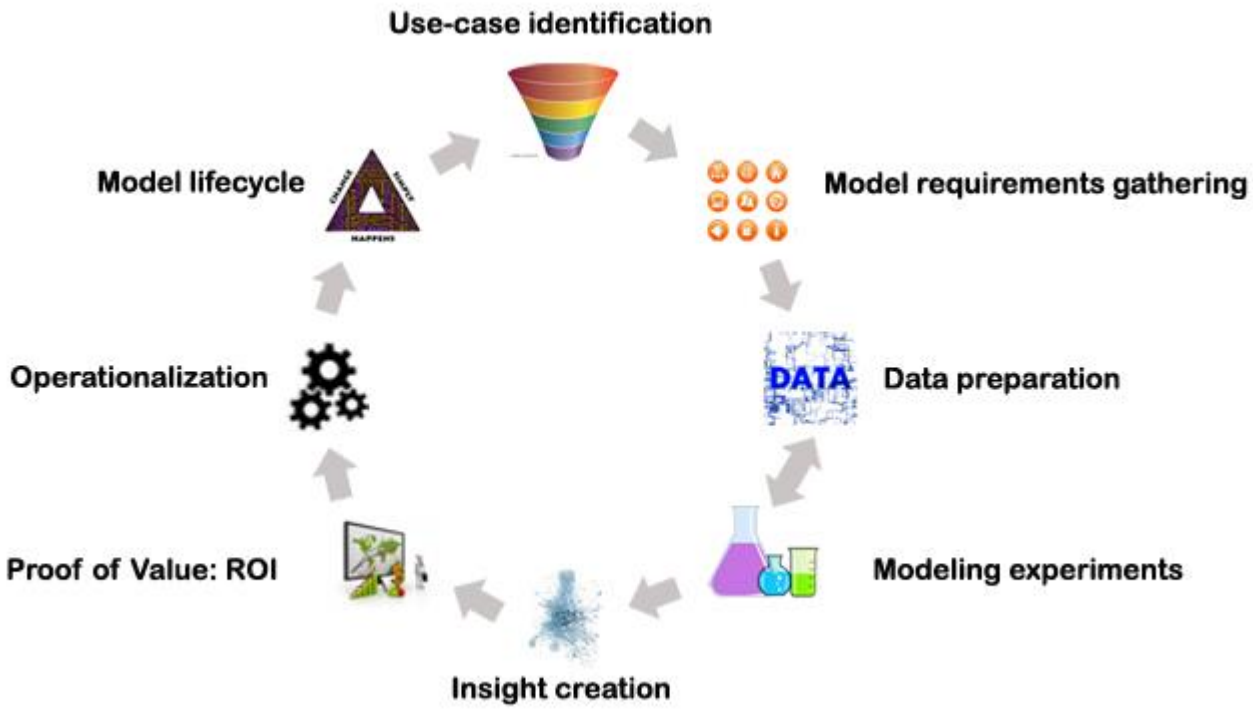
https://www.informit.com/articles/article.aspx?p=2473128&seqNum=11

SEMMA by SAS Institute

- Sample
  - Sampling the data by extracting a portion of a large data set big enough to contain the significant information, yet small enough to manipulate quickly.
  - Optional stage
- Explore
  - Exploration of the data by searching for unanticipated trends and anomalies in order to gain understanding and ideas.
- Modify
  - Modification of the data by creating, selecting, and transforming the variables to focus the model selection process.
- Model
  - Modeling the data by allowing the software to search automatically for a combination of data that reliably predicts a desired outcome.
- Assess
  - Assessing the data by evaluating the usefulness and reliability of the findings from the data mining process and estimate how well it performs.



Standard Methodology for Analytical Models



https://www.datascienceassn.org/content/standard-methodology-analytical-models

Standard Methodology for Analytical Models

Table 1

Phase	Description
Use-case identification	Selection of the ideal approach from a list of candidates
Model requirements gathering	Understanding the conditions required for the model to function
Data preparation	Getting the data ready for the modeling
Modeling experiments	Scientific experimentation to solve the business question
Insight creation	Visualization and dashboarding to provide insight
Proof of Value: ROI	Running the model in a small scale setting to prove the value
Operationalization	Embedding the analytical model in operational systems
Model lifecycle	Governance around model lifetime and refresh

https://www.datascienceassn.org/content/standard-methodology-analytical-models

Dataops

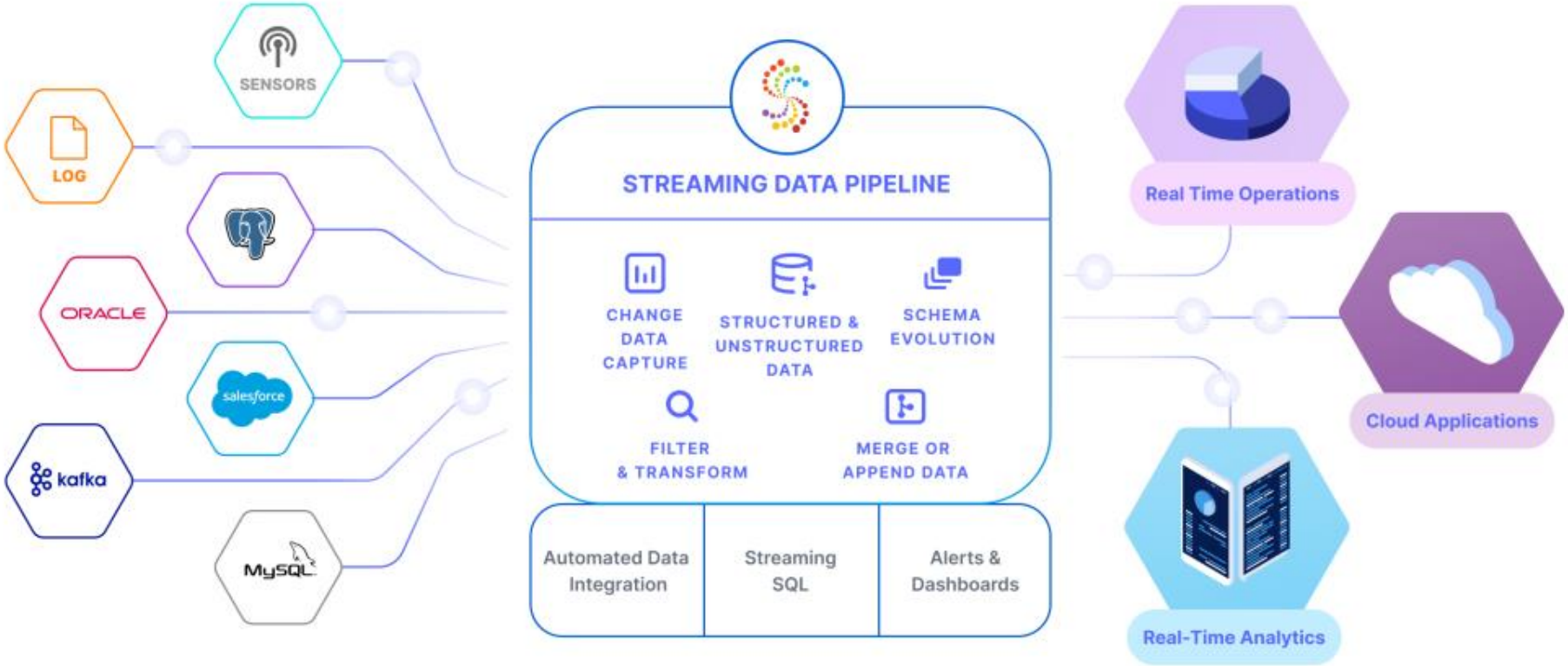
Dataops is a methodology that focuses on the integration and automation of data pipelines. It's all about streamlining the process of collecting, processing, and analyzing data and making it more efficient.

By automating processes and integrating data pipelines, companies can reduce errors, improve efficiency, and make better decisions.

Devops vs Dataops:
https://www.techtarget.com/searchbusinessanalytics/feature/Key-differences-in-uses-of-DataOps-vs-DevOps

https://www.thedataops.org/what-is-the-dataops-definition/

Dataops



https://www.thedataops.org/what-is-the-dataops-definition/

DevOps versus MLOps

DevOps (agile approach) provides benefits of shortening the development cycles, increasing deployment velocity, and dependable releases. To achieve these benefits, you introduce two concepts in the software system development:
Continuous Integration (CI) and Continuous Delivery (CD)

ML system is a software system, so similar practices help guarantee that you can reliably build and operate ML systems at scale
In ML, there are a few notable differences:

- CI is no longer only about testing and validating code and components, but also testing and validating data, data schemas, and models.
- CD is no longer about a single software package or a service, but a system (an ML training pipeline) that should automatically deploy another service (model prediction service).
- CT (Continuous Training) is a new property, unique to ML systems, that's concerned with automatically retraining and serving the models.

https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning

Text Books

	Author(s), Title, Edition, Publishing House
T1	Tan P. N., Steinbach M & Kumar V. “Introduction to Data Mining” Pearson Education
T2	Introducing Data Science by Cielen, Meysman and Ali
	Public Sources indicated in slides

Thank You